a mirror-smooth surface without any blemishes.

These wafers are then cleaned with pure water and then a photoresist coating is evenly applied on the surface. This photoresist coating when exposed to Ultraviolet (UV) light becomes water soluble. Remember that logic is designed across multiple layers and using controlled exposure through a stencil, the first layer of the CPU is etched onto the photoresist surface. The exposed areas are then dissolved and the another layer of photoresist is applied. Another controlled UV exposure takes place to etch the second layer of the CPU onto the silicon. The wafer is washed again. By repeating the process of adding photoresist and then exposing the wafer to UV, the individual layers of the IC are etched onto the silicon.

At the end of the process, all remaining photoresist layers are cleaned off and your CPU is completely etched. At this stage, you only have one type of material on the entire wafer. Electric current is not going to move from point A to point B on a circuit if it's entirely made of the same material. In order to

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Logic Chip

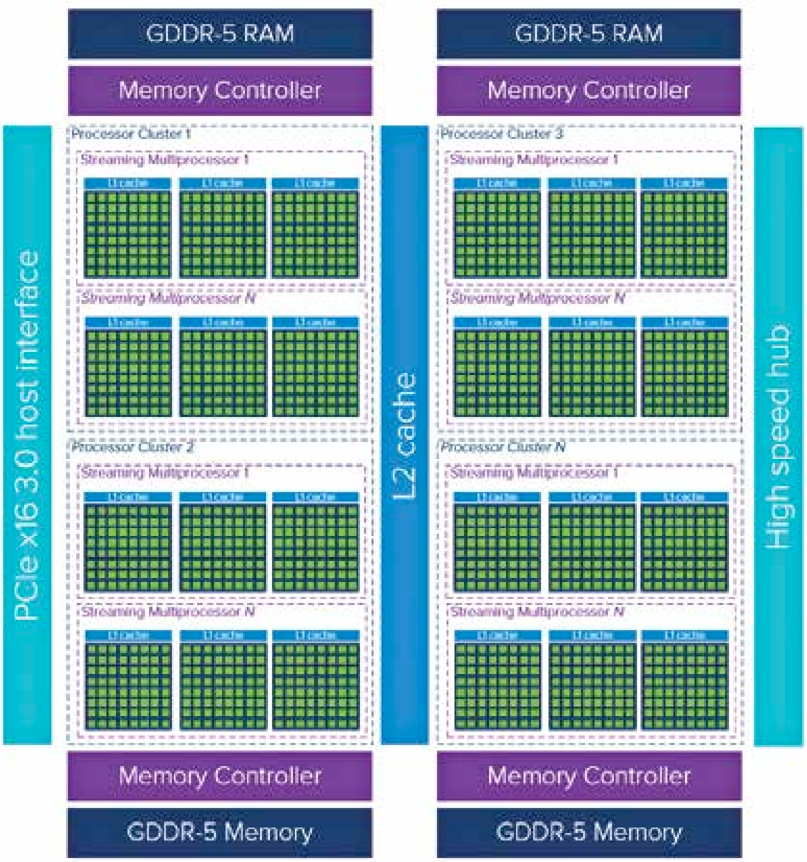
Logic chip are an umbrella term for processing chips, and can be differentiated from memory chips that are used for storage.

Microarchitecture

The microarchitecture refers to the specific way in which an instruction set is realised in a particular processor, known as an implementation. The same instruction set may have multiple implementations.

Microprocessor

Also known as the processor or the CPU, this is the component of a machine that executes the instructions provided by the OS or applications.



Credit: Niels Hagoort

More cores = more compute

bring about a difference, the wafer is then bombarded with ions. All exposed surfaced that get bombarded, will behave electrically different. The whole iterations of using photoresist coatings and creating different layers is repeated again. You finally have the CPU completely etched onto a wafer.

The only issue is that hundreds of CPUs are etched on a single wafer in a gridlike manner. And there are differences between two CPUs that are side by side on the wafer. This is primarily due to impurities in the silicon. So if the wafer was for a high-end processor, some of the CPUs, due to contamination, will perform at a reduced level. This is gauged by connecting hundreds of thousands of tiny probes which touch upon various points of the wafer and run tests.

A CPU which performs the best in this testing ends up being labelled as the top SKU. In the case of Intel, it would be a Core i9 processor and in the case of AMD it would be a Ryzen 7 processor. A CPU from the same wafer which doesn't perform just as good will be labelled a mid-range SKU such as a Core i5 or a Ryzen 5 and so on. This process is called binning, and it helps manufacturers identify the best chips from a wafer.

Once this is done, the wafer is cut up to reveal each individual processor. It is then connected to a substrate (PCB) onto which the silicon chip is affixed. Atop of this, they'll add a thermal interface material (TIM) such as solder or thermal paste. And then they'll seal the processors off with a metal heat-sink. Your CPU is now ready. All that remains is for you to find a compatible motherboard and you can start playing the latest games or whichever app you desire.

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